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Date: 2026-10-27
To: Clio Vega (cc Robin Langer)
Title: Q45: the weighted peel. Q30 without Buch–Ravikumar via T-equivariant independence
Work-in-progress commit: `76c28bfc00a1ffc7a93ec8cd30e69ec6fe63668e`
Full proof: `proofs/2026-10-27-weighted-peel.md` in `rick-research` at commit `0d01164615f9`, `https://github.com/grandpa-rick/rick-research/blob/0d01164615f91f1a0b6c03801a2602ed51fb512a/proofs/2026-10-27-weighted-peel.md`
Scripts: `scripts/prove259/` (`verify.py`, `cmp_q45.py`, outputs) in `work-in-progress` at the commit above.
Registry nodes: `q488-Q45-equivariant-peel-lift`, `q488-Q45-strict-character`, `q488-Q45-Q30-BR-free`.
Status: post-submission material. It is not for the FPSAC cut and the draft is not touched.

Summary

I give each shifted box a weight β. Along a row the weights telescope, so the LS endpoint $\pi(1)$ is linear in the order-preserving map. That yields T-characters for both the LS-parity model and its strict (reciprocal) side. The peel lemma `lem:peel` preserves these weights. The T-characters ch_ρ are linearly independent, although the Hilbert polynomials H_ρ are not (already at $\kappa = (3,1)$). So the equivariant hyperplane recursion pins down the Pieri coefficients, and forgetting T gives Theorem Q30 (Thm 4.8) *without (BR).*

Setting

We work in type C_m with ε-coordinates, $\omega = \omega_m = \varepsilon_1 + \cdots + \varepsilon_m$ and $x^\mu = x_1^{\mu_1} \cdots x_m^{\mu_m}$. Let κ be strict with parts $\le m$. Write $D(\kappa)$ for the shifted diagram and $P_\kappa = D(\kappa)$ with the product order. Let $w_\kappa \in W^P$ satisfy $w_\kappa \omega = v_\kappa$, which has -1 exactly at the positions $m+1-p$, $p \in \kappa$. Put $\mathrm{ch}_\kappa(j) := D_{w_\kappa}(e^{j\omega})$ for $j \in \mathbb{Z}$ (Demazure operators) and $n_\kappa(j) := (-1)^{|\kappa|}\mathrm{ch}_\kappa(-j)$. For $\rho \subseteq \kappa$ let $\theta = D(\kappa) \setminus D(\rho)$. A *V-map* on θ is a strictly order-preserving $g : \theta \to \{e < o\}$ with $g = e$ on diagonal boxes, and $V(\theta)$ is the number of V-maps.

Box weights. For a box $x = (a, b)$ with offset $d = b - a$: $\beta(x) = \varepsilon_m$ if $d = 0$, and $\beta(x) = \varepsilon_{m-d} - \varepsilon_{m+1-d}$ if $1 \le d \le m-1$.

Lemma 0 (telescoping). $\sum_{x \in D(\tau)} \beta(x) = \sum_{p \in \tau} \varepsilon_{m+1-p}$, hence $w_\tau \omega = \omega - 2\sum_{x \in D(\tau)} \beta(x)$. For the pairing $\ell(\mu) = \sum_c (m+1-c)\mu_c$ we have $\ell(\beta(x)) = 1$ for every box.

Statements (trust grade in brackets)

Theorem A (T-character of the LS-parity model) [proved, given (LS)]. For $j \ge 0$,

$$\mathrm{ch}_\kappa(j) = \sum_G x^{\,j\, w_\kappa \omega + \sum_x G(x)\beta(x)},$$

where the sum runs over order-preserving $G : P_\kappa \to \{0, \ldots, 2j\}$ that are even on diagonal boxes.

Theorem B (strict side) [proved, given (LS) and Stanley's cone reciprocity]. For $j \geq 1$,

$$n_\kappa(j) = (-1)^{|\kappa|}\mathrm{ch}_\kappa(-j) = \sum_f x^{-j\, w_\kappa\omega - \sum_x f(x)\beta(x)},$$

where the sum runs over strictly order-preserving $f : P_\kappa \to \{1, \ldots, 2j-1\}$ that are even on diagonal boxes. At $x = 1$ this is `thm:main`(b).

Theorem C (the weighted peel) [proved]. For $j \geq 1$ and every κ,

$$n_\kappa(j+1) = \sum_{\rho\subseteq\kappa} P_{\kappa\rho}(x)\, n_\rho(j), \qquad P_{\kappa\rho}(x) := \sum_{g \text{ V-map on } \theta} x^{-w_\kappa\omega - \sum_{x\in\theta,\, g(x)=o}\beta(x)}.$$

The bijection of `lem:peel` preserves the weights of Theorem B. In particular $P(x)$ is coefficient-positive, $P(1) = I + V$, $\mathrm{supp}\, P = \mathrm{supp}(I+V)$, and each entry is a sum of exactly $V(\theta)$ monomials. Moreover, for every $i \in \mathbb{Z}$,

$$\mathrm{ch}(i-1) = S\, P(x)\, S \cdot \mathrm{ch}(i), \qquad S = \mathrm{diag}((-1)^{|\kappa|}). \tag{$\dagger$}$$

Equivalently, the Wake 259 linear solve $C(x)$ (with $\mathrm{ch}(j+1) = C\,\mathrm{ch}(j)$) equals $SP^{-1}S$.

Lemma D (equivariant independence) [proved]. For any finite set R of strict partitions with parts $\leq m$, the functions $j \mapsto \mathrm{ch}_\rho(j)$ $(j \geq 1)$, $\rho \in R$, are linearly independent over $\mathbb{Q}(x)$. *Idea:* grade by ℓ. The lowest monomial of $\mathrm{ch}_\rho(j)$ is $x^{jw_\rho\omega}$, of degree $j\ell(\omega) - 2j|\rho|$. Isolate the maximal $|\rho|$ with minimal δ_ρ, and apply Vandermonde to the distinct $x^{w_\rho\omega}$. This is exactly what fails at $x = 1$, where the H_ρ are dependent for $\kappa = (3,1)$.

Theorem E (equivariant K-Chevalley for L^{-1} on LG; Q30 without (BR)) [proved modulo (KT), a standard Thomason input whose locator is not yet written out]. The inputs are (G0), (G1) in its T-equivariant (character) form, and (KT): the image of $K_0^T(X_\kappa)$ in $K_T(X)$ is spanned over $R(T)$ by $[\mathcal{O}_{X_\rho}]_T$, $\rho \subseteq \kappa$. In $K_T(LG(m,2m))$,

$$[\mathcal{O}_{X_\kappa} \otimes L^{-1}]_T = \sum_{\rho\subseteq\kappa} A_{\kappa\rho}[\mathcal{O}_{X_\rho}]_T, \qquad A_{\kappa\rho} = \psi\big((-1)^{|\theta|}P_{\kappa\rho}(x)\big),$$

where ψ is the duality $x \mapsto x^{-1}$ of $R(T)$. So A is unique, and $A_{\kappa\rho}$ is $(-1)^{|\theta|}$ times a positive sum of $V(\theta)$ monomials. Forgetting T gives

$$[\mathcal{O}_{X_\kappa}] \cdot \mathcal{O}^1 = \sum_{\rho\subsetneq\kappa} (-1)^{|\theta|-1} V(\theta)\, [\mathcal{O}_{X_\rho}],$$

which is Theorem `thm:Q30` ($c_\rho = (-1)^{|\theta|-1}V(\theta)$). It is proved from (G0), (G1), (LS), (KT) and Stanley reciprocity, **without (BR)**. *Proof sketch:* use (KT) to write the expansion, and apply $\Phi(F)(j) = \psi\chi_T(X, F \otimes L^j)$ to get $\mathrm{ch}_\kappa(j-1) = \sum_\rho \psi(A_{\kappa\rho})\mathrm{ch}_\rho(j)$. Compare with (†), and Lemma D forces $\psi(A_{\kappa\rho}) = (-1)^{|\kappa|+|\rho|}P_{\kappa\rho}$.

Checks

`verify.py` builds $\mathrm{ch}_\kappa(j)$ by Demazure operators. Theorems A, B and C hold at the character level for $m = 2$ $(j \leq 4)$, $m = 3$ $(j \leq 4)$, $m = 4$ $(j \leq 5)$ and $m = 5$ $(j \leq 4)$. All 22 nonzero entries of the combinatorial P at $m = 3$ equal the Wake 259 linear solve $SC^{-1}S$. Negative control: flipping β on offset-1 boxes to $\varepsilon_{m-1} + \varepsilon_m$ makes A, B and C all fail. Sanity example ($m = 3$): $P_{(2),(1)} = (x_2 + x_3)/(x_1x_3)$.

Gaps and ownership

Theorems A–C and Lemma D are complete given (LS) and Stanley's cone reciprocity (Beck–Robins Thm 4.3, cited by statement). Theorem E additionally uses (KT). (KT) is the equivariant version of the localization argument in `lem:hyperplane`, but the Thomason citation with a locator is not yet written out. *Owner caveat:* the equivariant Chevalley formula is owned (Lenart–Shimozono 1203.3237 Thm 3.2/Cor 3.4; Lenart–Postnikov math/0309207; Pittie–Ram). By uniqueness its ω_m case must equal A. I have not checked that it reads literally as P. What is new is the route (LS $\to$ Stanley reciprocity $\to$ strict characters $\to$ weighted peel $\to$ independence $\Rightarrow$ Pieri) and the closed shifted-diagram form, with one monomial per V-map.

Ask (not urgent, post-FPSAC). Clio: if you have bandwidth, please check Lemma D and Theorem E.